by an interference pattern. Celluloid, glass, bakelite etc. if they are free from strain, do not exhibit the property of birefringence. But these materials exhibit the property of double refraction temporarily when subjected to internal stress.

If such a strained material is placed between two crossed nicols, an interference pattern is observed. The photograph of the pattern is taken and the stress at various points is studied. The stress is maximum at points where the interference pattern is close. The stress produced in a girder when loaded is estimated by this method. The model of the girder made of a transparent plastic material is placed between the crossed nicols and then loaded. The stresses produced at different positions can be analysed and estimated. This method of stress analysis is commonly used in preference to other methods.

12.32 FIBRE OPTICS

When a beam of light enters one end of a transparent rod, say a gloss rod, the light beam is totally internally reflected and gets trapped within the rod. Such a rod is called a light pipe.

A bundle of fine fibres also behaves in the same manner. The bundle has the advantage of flexibility. A beam entering at one end can be transmitted at the other end through the fibre, even though the fibre is curved.

A bundle may consist of thousands of individual fibres with diameter in the range 0.0002 cm to 0.001 cm. In case the fibres are arranged so that the free ends on both sides are at the same relative positions, the bundle can transmit an image from one end to the other end.

The study of the properties of such a bundle of fibres is known as **fibre optics**. Applications of fibre optics in medicine are wide ranging.

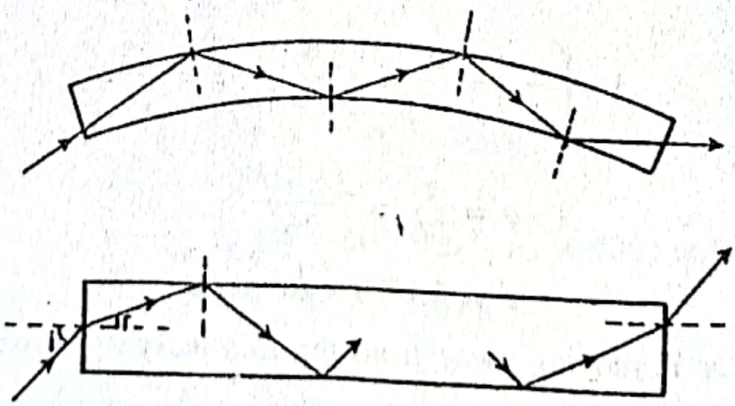

Fig. 12.35

With the help of fibre optics, it is possible to study the interior of the lungs and the other parts of the human body that cannot be viewed directly.

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It is also useful in the study of tissues and blood vessels far below the skin. For this purpose, a bundle of fibres is enclosed in a hypodermic needle. This flexible fiber bundles are used in medicine as *endoscopes*. Endoscopes are employed to make observations at a place that can be approached only along a curved path.

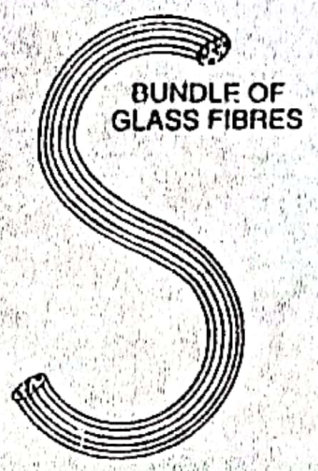

Fig. 12.36

Fibre bundles that are referred to above are said to be passive *i.e.* they merely convey light from one end to the other end. Recently **Snitzer** (1961) has produced active scintillating fibres in which light is actually generated and conveyed to the other end. Fibre lasers are useful to develop flexible high intensity laser probes.

12.33 LASER

Recent researches in solid state physics have revealed a new device called 'Laser' which means 'Light Amplification by Stimulated Emission of Radiation'. Towns and Schawlow in 1958 worked out the principle of Laser. The principle of a laser is similar to that of 'Maser' which means Micro wave Amplification by Stimulated Emission of Radiation. Due to this reason laser is also called 'Optical Maser'.

Laser works on the principle of quantum theory of radiation. When an electron in an orbit absorbs energy, it goes to an orbit of higher energy level. When the electron jumps back to a stable orbit, it emits radiations in the form of electro-magnetic waves. The energy of each photon = $h\nu$.

Laser consists of a ruby rod surrounded by a flash tube. One end of the ruby is highly silvered and the other end is semi-transparent. The flash tube is called the pump and it surrounds the ruby rod in the form of a spiral as shown in Fig. 12.37.

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Synthetic ruby consists of a crystal of Al O (aluminium oxide) in which a few of the aluminium atoms are replaced by chromium atoms. Chromium atoms have the property of absorbing green light. The electrons are excited to higher energy levels. When the electrons jump back to their original levels, red light is emitted. This is the reason why the ruby has

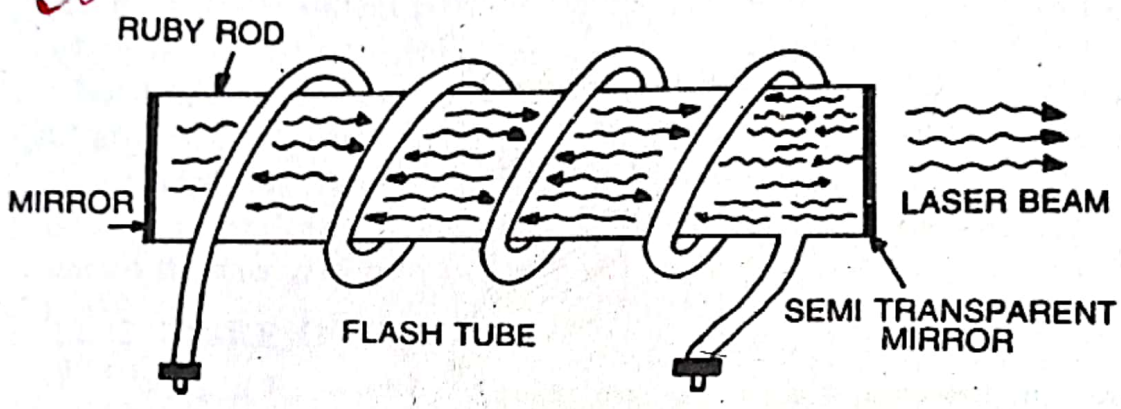

Fig. 12.37

natural red colour. The excited electrons are highly unstable in the higher energy levels and tend to give up their extra energy. *When these electrons in the excited state are exposed to the radiations of the same frequency which they are about to emit, the emission process is triggered instantaneously.* These radiations are exactly in phase with the exposed radiations and reinforce with one another.

These radiations are allowed to pass back and forth millions of times in the ruby with the help of the mirror at the two ends. A chain reaction is started and a powerful coherent beam of red light is obtained. This powerful beam of red light emerges out of the semi-transparent mirror at one end. Recently, a solid state injection laser has been invented. This laser can be turned on and off as desired.

Uses. Laser has a number of practical applications. It is useful in surgery, radar, welding microscopic wires and vaporizing any known material. A focussed laser beam has helped in the study of the moon's surface. It will provide a very handy communication medium because of its large band width. It can accommodate 80 million television channels. It is hoped that laser will be a useful tool in the study of distant planets.

12.34 SPONTANEOUS AND STIMULATED (INDUCED) EMISSION AND EINSTEIN'S A AND B COEFFICIENTS

An atom in the excited state emits radiations. This is possible in two ways, *viz.*

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(*i*) Spontaneous emission and (*ii*) stimulated emission. Spontaneous emission is not caused by any external effect. Stimulated or induced emission is due to the presence of electromagnetic radiation of suitable frequency. One of the main features associated with stimulated emission is the fact that the emitted radiation has the same frequency, phase and polarisation as the inducing radiation. This was predicted by Einstein in 1917 purely from thermodynamical considerations. Further, the emitted stimulated radiation propagates in the same direction as the inducing radiation. This has led to the development of laser technology recently.

Consider an atom in two states. Let the number of atoms per unit volume per unit time in state 1 be N_1 and in state 2, N_2. Let ψ represent the energy density of radiation for frequency ν. Then $\psi\, d\nu$ represents the energy density for radiation between frequency range ν and $\nu + d\nu$. Assuming that the rate of absorption of radiation of frequency ν from state 1 to state 2 is proportional to the radiation energy density ψ, the number of absorptions per unit volume per unit time will be given by

$$N_1 B_{12} \psi$$

Here B_{12} is a constant of proportionality from state 1 to state 2.

The number of spontaneous emissions per unit volume per unit time is given by

$$N_2 A_{21}$$

The number of stimulated emissions per unit-volume per unit time depends upon N_2 and ψ.

Therefore, the number of stimulated emissions per unit volume per unit time is given by $N_2 B_{21}\, \psi$

When thermal equilibrium is reached,

$$N_1 B_{12} \psi = N_2 A_{21} + N_2 B_{21} \psi \qquad \text{...(i)}$$

or

$$\psi = \frac{N_2 A_{21}}{N_1 B_{12} - N_2 B_{21}}$$

$$\psi = \frac{A_{21}}{(N_1 / N_2) B_{12} - B_{21}} \qquad \text{...(ii)}$$

According to Boltzmann's law, at thermal equilibrium

$$N_2 = N_1 e^{-\left[\frac{E_2 - E_1}{kT}\right]}$$

$$\frac{N_1}{N_2} = e^{\left[\frac{E_2 - E_1}{kT}\right]} \qquad \text{...(iii)}$$

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Substituting this value in equation (*ii*)

$$\psi = \frac{A_{21}}{\left[e^{\left[\frac{E_2 - E_1}{kT}\right]}\right] B_{12} - B_{21}} \quad ...(iv)$$

According to Planck's law of radiation,

$$\psi = \left[\frac{8\pi h\nu^2}{c^3}\right]\left[\frac{1}{\left(e^{\frac{h\nu}{kT}}\right) - 1}\right] \quad ...(v)$$

Taking $E_2 - E_1 = h\nu$ and rearranging equation (*iv*), we get

$$\psi = \left(\frac{A_{21}}{B_{12}}\right)\left[\frac{1}{(e^{h\nu/kT}) - \frac{B_{21}}{B_{12}}}\right] \quad ...(vi)$$

Comparing equations (*v*) and (*vi*),

$$\frac{B_{21}}{B_{12}} = 1 \text{ or } B_{21} = B_{12} \quad ...(vii)$$

and

$$\frac{A_{21}}{B_{12}} = \frac{8\pi h\nu^3}{c^3}$$

or

$$A_{21} = (B_{12})\left(\frac{8\pi\, h\nu^3}{c^3}\right) \quad ...(viii)$$

or

$$A_{21} = (B_{21})\left[\frac{8\pi\, h\nu^3}{c^3}\right] \quad ...(ix)$$

Equations (*vii*) and (*ix*) are Einstein's equations. A_{21}, B_{12} and B_{21} are called Einstein's coefficients. These coefficients are related to one another.

12.35 POPULATION INVERSION AND HELIUM-NEON LASER

Let N_1 be the number of atoms per unit volume per unit time in state 1 (ground state) and N_2 the number of atoms per unit volume per unit time in state 2 (excited state). If $N_1 > N_2$, it means that the incident beam is absorbed and the energy is used in exciting the atoms from state 1 to state 2. However, if $N_2 > N_1$, the beam gets amplified. The phenomenon is called light amplification. As the number of atoms per unit volume per unit time in the excited state is higher than the ground state, it is called **population inversion.** For stimulated emission, population inversion is a necessity.

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Helium-Neon laser consists of a narrow discharge tube filled with a mixture of Helium and Neon in the ratio 10 : 1. The pressure inside the tube is maintained at 1 mm of Hg. M_1 and M_2 are two mirrors (plane or concave) kept at the ends of the tube. Mirror M_1 is highly silvered and M_2 is partially transparent. M_1 and M_2 together constitute a resonating

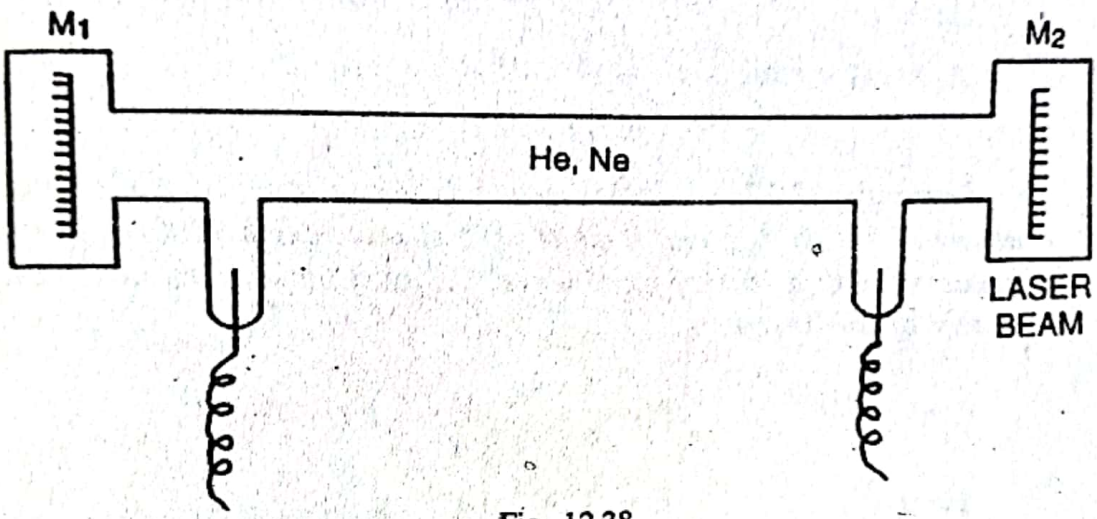

Fig. 12.38

system. When the electrodes are connected to a source of high potential, electric discharge passes. Helium atoms are excited first. Due to the collision of helium atoms with neon atoms they are excited. These collisions result in population inversion. As soon as population inversion is reached, any of the spontaneously emitted photons will trigger laser action in the tube. A well collimated monochromatic intense beam emerges out of the tube.

Gas lasers are preferred compared to Ruby lasers because the laser beam is more directional and more monochromatic. The defects due to crystalline imperfections, thermal distortion and scattering are absent in gas lasers. Moreover, gas lasers can be operated continuously and there is no need for frequent cooling as in the case of solid state laser.

Example 12.11. *A laser beam has a wavelength of* 8×10^{-7} m *and aperture* 5×10^{-3} m. *The laser beam is sent to moon. The distance of the moon is* 4×10^5 km *from the earth. Calculate (i) the angular spread of the beam and (ii) the axial spread when it reaches the moon.*

(*i*) Let $d\theta$ be the angular spread

$$d\theta = \frac{\lambda}{d}$$

Here

$$\lambda = 8 \times 10^{-7} \text{ m}$$

$$d = 5 \times 10^{-3} \text{ m}$$

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$$d\theta = \frac{8 \times 10^{-7}}{5 \times 10^{-3}}$$

$$= 1.6 \times 10^{-4} \text{ radian}$$

(2) Areal spread $= (D\, d\theta)^2$

Here $D = 4 \times 10^5 \text{ km} = 4 \times 10^8 \text{ m}$

∴ Areal spread $= (4 \times 10^8 \times 1.6 \times 10^{-4})^2$

$$= \mathbf{4.096 \times 10^9 \text{ m}^2}$$

Example 12.12. *A laser beam has a power of 50* mW. *It has an aperture of* 5×10^{-3} m *and it emits light of wavelength 7200* Å. *The beam is focussed with a lens of focal length 0.1* m. *Calculate the area and the intensity of the image.*

$$d\theta = \frac{\lambda}{d}$$

Here $\lambda = 7200 \times 10^{-10}$ m

$d = 5 \times 10^{-3}$ m

∴ $$d\theta = \frac{7200 \times 10^{-10}}{5 \times 10^{-3}}$$

$$d\theta = 1.44 \times 10^{-4} \text{ radian}$$

(*i*) Areal spread $= [(d\theta) f]^2$

Here $f = 0.1$ m

∴ Areal spread $= (1.44 \times 10^{-4} \times 0.1)^2$

$$= 2.074 \times 10^{-10} \text{ m}^2$$

(*ii*) Intensity $= \dfrac{\text{Power}}{\text{Area}}$

$$I = \frac{\text{Power}}{\text{Areal spread}}$$

Here, Power $= 50 \times 10^{-3}$ watts

Areal spread $= 2.074 \times 10^{-10} \text{ m}^2$

∴ $$I = \frac{50 \times 10^{-3}}{2.074 \times 10^{-10}}$$

$$I = 2.411 \times 10^8 \text{ watts/m}^2.$$

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Example 12.13. *The coherence length for sodium light is* 2.945×10^{-2} m. *The wavelength of sodium light is 5890 Å. Calculate (i) the number of oscillations corresponding to the coherence length and (ii) the coherence time.*

Here, $\lambda = 5890 \times 10^{-10}$ m

$c = 3 \times 10^{8}$ m/s

Coherence length $L = 2.945 \times 10^{-2}$ m

(*i*) Number of oscillations in length *L*,

$$n = \frac{L}{\lambda}$$

$$= \frac{2.945 \times 10^{-2}}{5890 \times 10^{-10}}$$

$$= 5 \times 10^{4}$$

(*ii*) Coherence time $= \dfrac{L}{c}$

$$= \frac{2.945 \times 10^{-2}}{3 \times 10^{8}}$$

$$= \mathbf{9.82 \times 10^{-11}\ s}$$

Example 12.14. *A laser beam* $\lambda = 6000$ Å *on earth is focussed by a lens (or mirror) of diameter 2 m on to a crater on the moon. The distance of the moon is* 4×10^{8} m. *How big is the spot on the moon ? Neglect the effect of earth's atmosphere.* [*IAS, 1979*]

Here $\lambda = 6000$ Å $= 6 \times 10^{-7}$ m

$d = 2$ m

$D = 4 \times 10^{8}$ m

(*i*) Angular spread

$$d\theta = \frac{\lambda}{d}$$

$$d\theta = \frac{6 \times 10^{-7}}{2} = \mathbf{3 \times 10^{-7}\ radian}$$

(*ii*) Areal spread *i.e.* area of the spot on the moon,

$$A = (D \times d\theta)^2$$

$$= (4 \times 10^{8} \times 3 \times 10^{-7})^2$$

$$A = \mathbf{14400\ m^2 = 1.44 \times 10^{4}\ m^2}$$

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